

SYNOPSIS

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Topic: Honeypot-Driven Threat Detection Pipeline.

Project Guidance: Prof. Jaynish Mandal.

Honeypot-Driven Threat Detection Pipeline.

Introduction to Project:

Organizations face constant, automated attacks from bots and hackers scanning the internet for weak servers. Most small-scale setups and student-level security projects rely only on theoretical detection methods and lack real attack data to learn from. This project deploys a honeypot , a decoy server made to look like a real, vulnerable Linux machine that attracts real attackers on the open internet. Every login attempt, command, and action taken by an attacker is logged in detail without any real system being at risk.

These raw logs are then processed by a custom-built Python enrichment layer, which adds context such as the attacker's location, whether their IP address is already known for malicious activity, and how severe the attempted attack is. This enriched data is fed into Wazuh, an open-source SIEM (Security Information and Event Management) platform, which displays the information as live dashboards and triggers alerts based on custom detection rules replicating how a real-world Security Operations Center (SOC) monitors and responds to threats.

Proposed system:

Deploys a live honeypot (Cowrie) on a cloud server to capture real, ongoing attack attempts from across the internet.

Automatically enriches raw attack logs with geolocation, IP reputation, and severity scoring using a custom Python script.

Feeds enriched data into a centralized Wazuh SIEM for real-time dashboards and alerting.

Uses custom-written detection rules to flag brute-force attempts, known malicious IPs, and suspicious command patterns.

Designed to be extended in later phases with automated response, such as auto-blocking repeat-offender IP addresses.

All logs, alerts, and analysis are documented and version-controlled on GitHub for transparency and review.

Existing System:

Most conventional setups depend on static, signature-based detection with no real threat intelligence behind them.

Raw security logs are noisy and lack context, making manual analysis slow and error-prone.

There is little to no centralized visibility into attacker behavior trends over time.

Detection and response are typically manual, with no automated enrichment or correlation of data.

Modules:

Honeypot Deployment: Cowrie simulates a vulnerable SSH/Telnet Linux server to attract and safely log real attacker activity.

Log Ingestion: Collects and forwards raw activity logs generated by the honeypot for processing.

Enrichment Engine: A custom Python module that parses raw logs and adds geolocation, IP reputation (via free threat intel APIs), and severity classification.

SIEM Integration: Wazuh ingests enriched data, generates real-time dashboards, and manages centralized alerting.

Detection Rules: Custom-written rules to flag brute-force attempts, known malicious IPs, and suspicious credential patterns.

Reporting & Visualization: Dashboards showing attack trends, top attacking countries, and most commonly attempted passwords.

Automated Response (Phase 2): Auto-blocks repeat-offender IP addresses via firewall rules based on threat score, extending the project in Year 3.

Hardware and software requirement:

|Hardware System:

Processor: Intel i5 (or equivalent), 2 GHz or higher

Memory: 8 GB RAM

Storage: 20–50 GB (Cloud VPS storage)

Graphics: No dedicated GPU required

Network: Stable internet connection

Hosting: Cloud VPS instance (e.g. Oracle Cloud Free Tier — ARM-based, free)

Software System:

Operating System: Ubuntu Server (Linux) on VPS; Linux distribution on development machine

Honeypot: Cowrie (open-source SSH/Telnet honeypot)

SIEM Platform: Wazuh (open-source security monitoring platform)

Scripting Language: Python 3

Threat Intelligence: AbuseIPDB API and IP geolocation APIs (free tier)

Version Control: Git and GitHub